APPLIED PHYSICS FOR CSE STREAM(BPHYS102/202)

MODULE- 2

QUANTUM MECHANICS

de Broglie HYPOTHESIS & UNCERTAINTY PRINCIPLE

Pre-requisite: Wave–Particle Dualism

Experiments show that:

- Light behaves as a **wave** (interference, diffraction)
- Light also behaves as a **particle** (photoelectric effect)

This dual behavior led to the idea that **matter particles may also exhibit wave nature**.

1. de Broglie Hypothesis

Statement

According to **de Broglie**, every moving material particle is associated with a wave called a **matter wave**.

The wavelength associated with a particle of momentum $p$is:

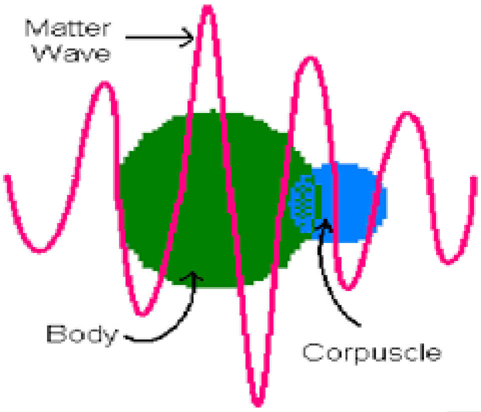
$$\boxed{\lambda = \frac{h}{p}}$$

For a particle of mass $m$and velocity v:

$$\boxed{\lambda = \frac{h}{mv}}$$

where
h= Planck's constant

Matter Wave Diagram



2. Derivation of de Broglie Wavelength (by Analogy)

For a photon:

$$E = h\nu \text{ and } E = pc$$

Equating,

$$pc = h\nu$$

Since $\nu = \frac{c}{\lambda}$,

$$p = \frac{h}{\lambda} \Rightarrow \boxed{\lambda = \frac{h}{p}}$$

For matter particles:

$$\lambda = \frac{h}{mv}$$

3. Phase Velocity and Group Velocity

3.1 Phase Velocity

Phase velocity is the velocity of a **constant phase point** of the wave.

$$\boxed{v_p = \frac{\omega}{k}}$$

For matter waves:

$$v_p = \frac{E}{p}$$

Phase velocity may exceed speed of light but carries **no physical information**.

3.2 Group Velocity

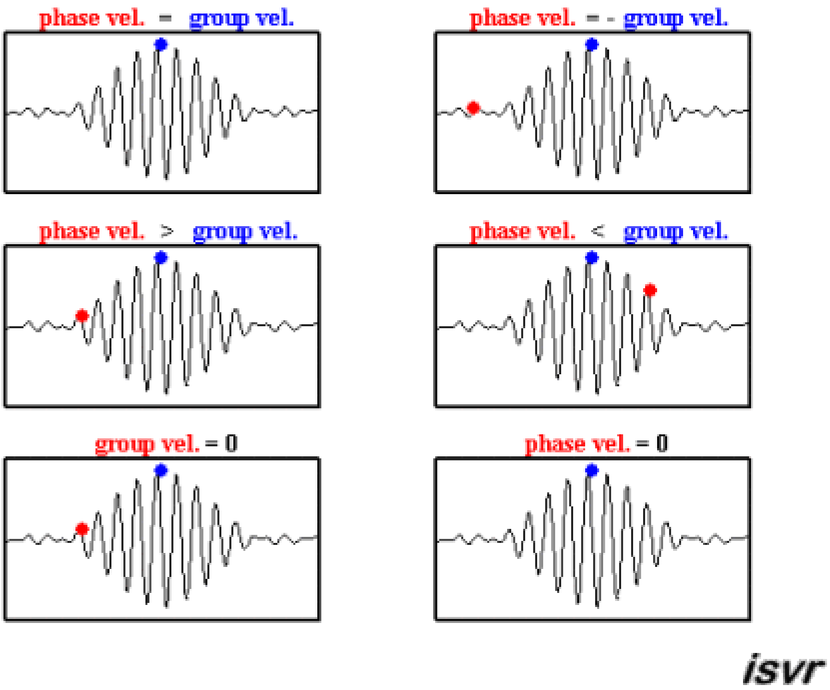
Group velocity represents the velocity of the **wave packet** and corresponds to the **particle velocity**.

$$\boxed{v_g = \frac{d\omega}{dk}}$$

For matter waves:

$$\boxed{v_g = v}$$

Phase & Group Velocity Diagram



4. Heisenberg's Uncertainty Principle

Statement

It is impossible to simultaneously determine the **exact position and momentum** of a particle.

$$\boxed{\Delta x\, \Delta p \geq \frac{h}{4\pi}}$$

Similarly,

$$\Delta E \, \Delta t \geq \frac{h}{4\pi}$$

5. Application of Uncertainty Principle

Non-existence of Electron Inside the Nucleus (Non-relativistic)

If an electron were confined inside a nucleus of radius:

$$\Delta x \approx 10^{-14} \text{ m}$$

Then,

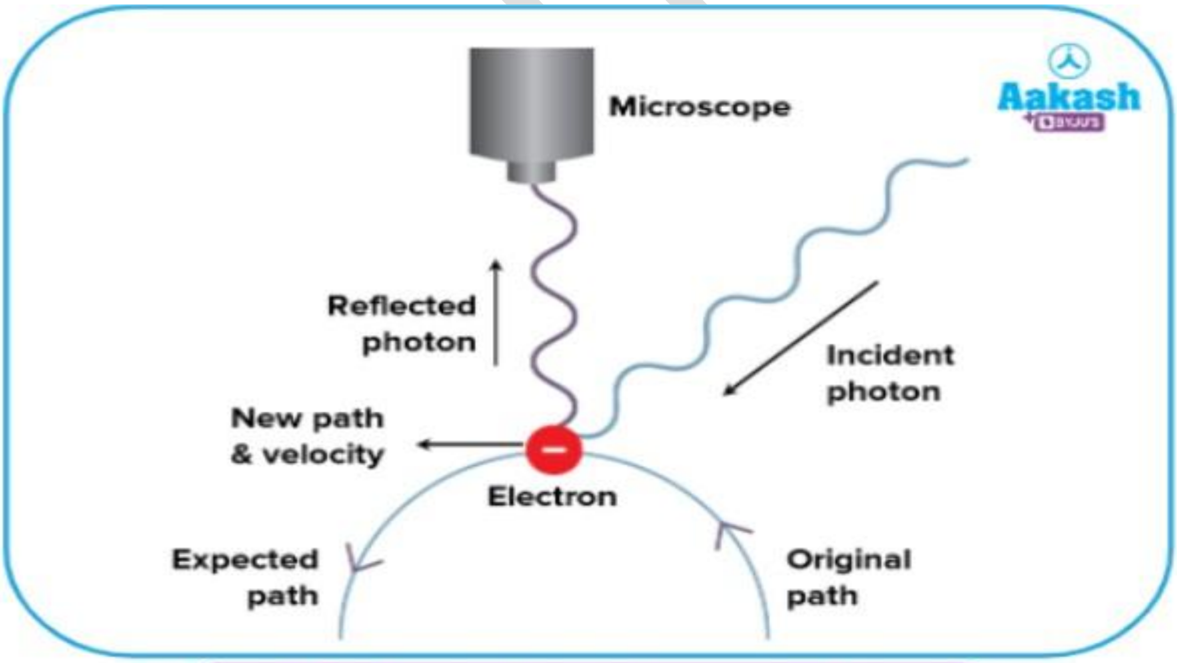
$$\Delta p \geq \frac{h}{4\pi \Delta x}$$

This gives an extremely large momentum and kinetic energy, which is **not observed experimentally**.

Hence:

$$\boxed{\text{Electron cannot exist inside the nucleus}}$$

Uncertainty Principle Diagram



Experimental setup for uncertainty measurement of position and momentum

6. Principle of Complementarity

According to **Bohr**:

- A quantum system exhibits **either wave nature or particle nature**
- Both cannot be observed simultaneously

- Both descriptions are necessary for complete understanding

WAVE FUNCTION & SCHRÖDINGER WAVE EQUATION

1. Wave Function

Definition

In quantum mechanics, the state of a particle is described by a complex function called the **wave function**, denoted by:

$$\psi(x,t)$$

It contains all the measurable information about the particle.

2. Physical Significance of Wave Function (Born Interpretation)

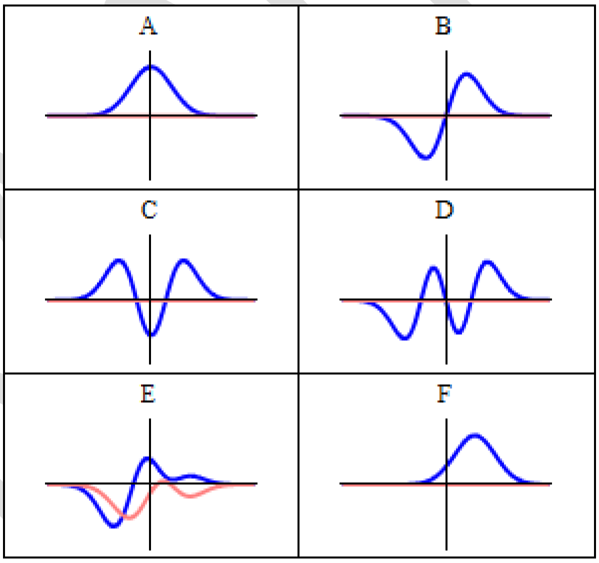
The wave function itself has **no direct physical meaning**.
According to **Born**, the measurable quantity is:

$$\boxed{|\psi(x,t)|^2}$$

which represents the **probability density** of finding the particle at position $x$at time t.

$$|\psi|^2\,dx = \text{Probability of finding particle between } x \text{ and } x+dx$$

Wave Function & Probability Density Diagram (REQUIRED)



3. Normalization of Wave Function

Since the particle must exist **somewhere in space**, total probability must be unity.

$$\boxed{\int_{-\infty}^{+\infty} |\psi|^2\,dx = 1}$$

This condition is called **normalization** of the wave function.

4. Expectation Value of a Physical Quantity

The **expectation value** of any observable Qrepresented by operator $\hat{Q}$is given by:

$$\boxed{\langle Q \rangle = \int \psi^* \hat{Q}\, \psi \, dx}$$

Examples:

Position:

$$\langle x \rangle = \int x \mid \psi \mid^2 \, dx$$

Momentum:

$$\langle p \rangle = \int \psi^* \left(\quad -i\hbar \frac{d}{dx} \right) \psi \, dx$$

5. Eigen Functions and Eigen Values

If an operator $\hat{Q}$acts on a function ψsuch that:

$$\boxed{\hat{Q}\psi = q\psi}$$

then:

- $\psi$is an **eigen function**
- $q$is the corresponding **eigen value**

Eigen values represent the **allowed measurable values** of the physical quantity.

6. Time Independent Schrödinger Wave Equation

Derivation (Final Form as per Syllabus)

Starting from the general Schrödinger equation and separating variables, we obtain the **time independent Schrödinger wave equation**:

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)}$$

where

$$\hbar = \frac{h}{2\pi}$$

$V(x)$= potential energy
E= total energy of the particle

This equation governs the **spatial behavior** of the particle.

7. Significance of Schrödinger Equation

- Determines allowed **energy states**
- Predicts **probability distribution**
- Explains atomic and sub-atomic behavior
- Foundation of quantum mechanics

PARTICLE IN A ONE-DIMENSIONAL INFINITE POTENTIAL WELL

1. One-Dimensional Infinite Potential Well (Particle in a Box)

Definition

A particle is confined to move in a region $0 \leq x \leq L$by infinitely high potential barriers at $x = 0$and $x = L$.

$$V(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & x \leq 0 \text{ or } x \geq L \end{cases}$$

Because the potential is infinite outside the box, the particle **cannot exist** there.

2. Schrödinger Equation Inside the Box

For $0 < x < L, V(x) = 0$.
The time-independent Schrödinger equation reduces to:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

or

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0, \text{where } k^2 = \frac{2mE}{\hbar^2}$$

3. Boundary Conditions

Because $V = \infty$at the walls:

$$\psi(0) = 0, \psi(L) = 0$$

These conditions determine the **allowed states**.

4. Allowed Wave Functions (Eigen Functions)

Solving with boundary conditions gives:

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), n = 1,2,3, \ldots}$$

Each $n$corresponds to a different stationary state

5. Quantization of Energy States

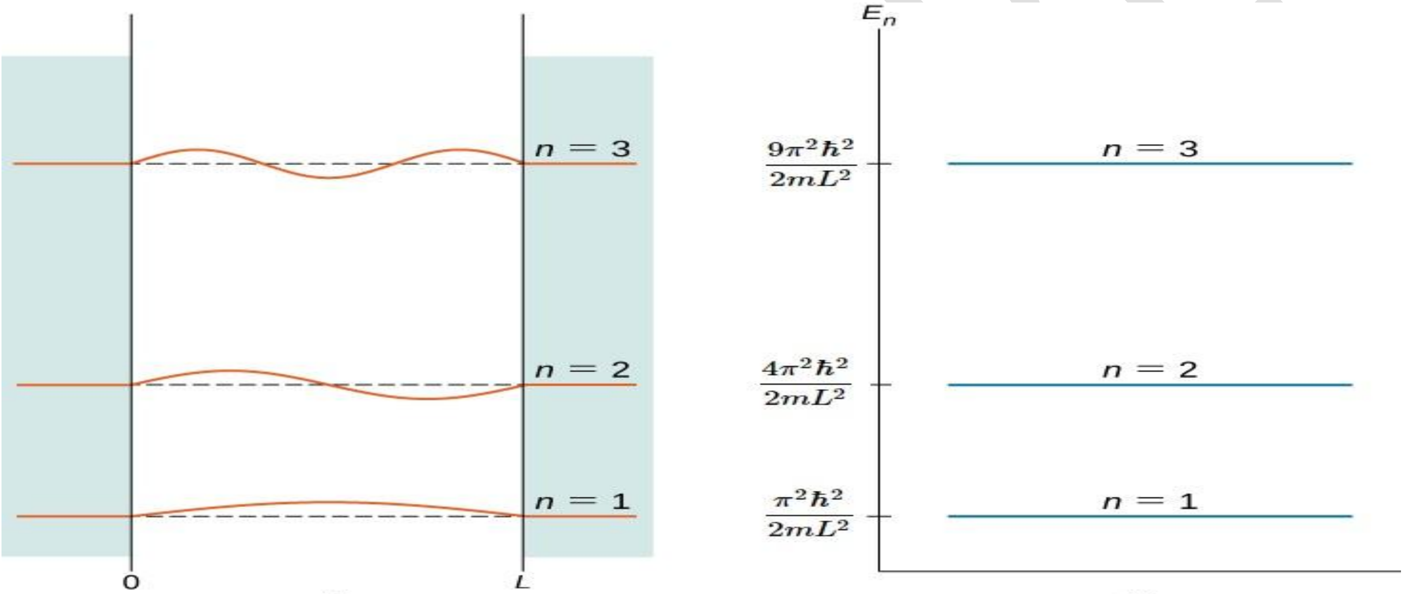
The allowed energies are:

$$\boxed{E_n = \frac{n^2 h^2}{8mL^2}, n = 1,2,3, \ldots}$$

Important Consequences

- Energy is **quantized** (discrete values only)
- **Ground state energy** ($n = 1$) is **non-zero**
- Energy increases as n^2

6. Energy Level Diagram



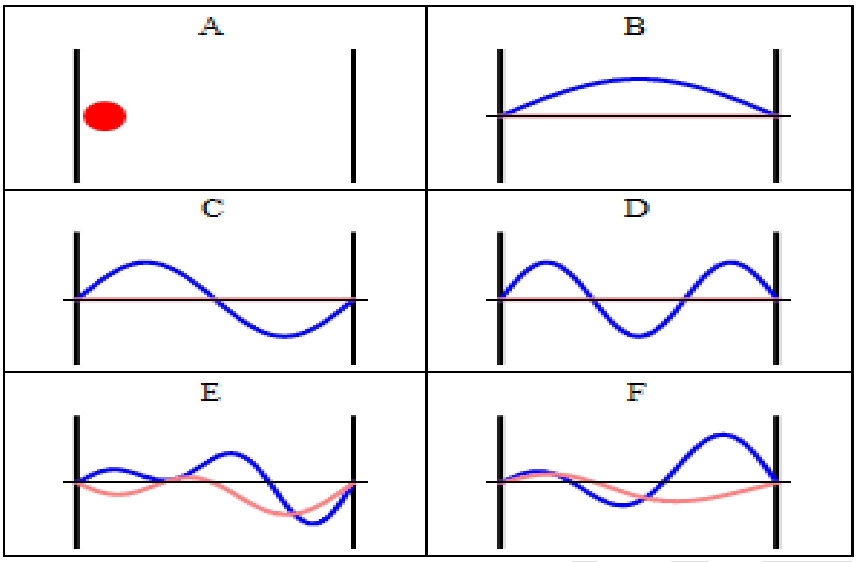
(a) (b)

7. Waveforms (Eigen Functions)

- $n = 1$: one half-wave
- $n = 2$: one full wave
- $n = 3$: one and a half waves

Number of nodes $= n - 1$.

Wavefunctions for n = 1, 2, 3



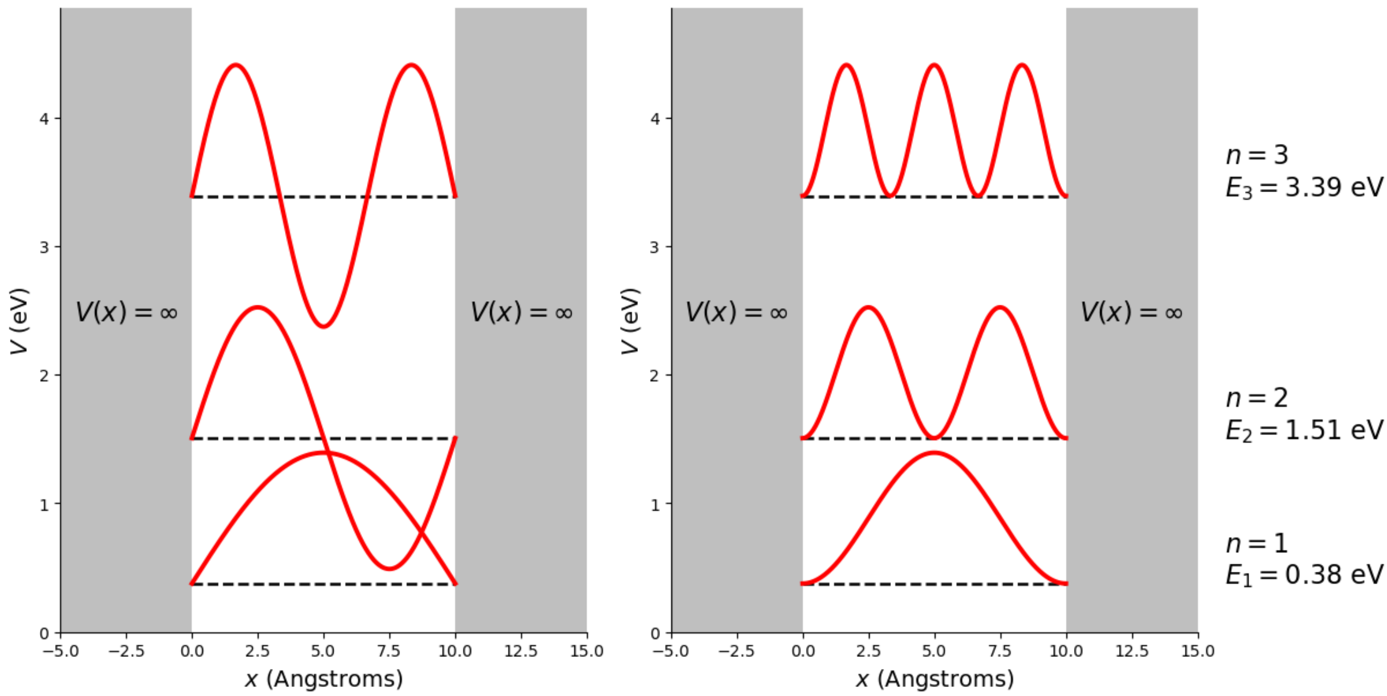
8. Probability Density

The probability density is:

$$\boxed{|\,\psi_n(x)\,|^2 = \frac{2}{L}\sin^2\left(\frac{n\pi x}{L}\right)}$$

It gives the likelihood of finding the particle at position x.

Probability Density Graphs



9. Expectation Value of Position

For the particle in a box:

$$\boxed{\langle x \rangle = \frac{L}{2}}$$

This means the particle is, on average, found at the **center of the box** in all stationary states.

10. Key Observations

- Discrete energy levels confirm **quantum behaviour**
- Zero-point energy exists
- Probability is not uniform
- Wave nature dominates microscopic systems